Idea Submission

Team Name - 100xDev
Theme - FinTech x AI Security
Project Name - Bio-Pulse Authenticator
Team Leader - Archie Yadav
College / Institute Name - Baderia Global Institute of Engineering and Management

Project Overview

The Problem & Bio-Pulse Solution

The Scary Reality: Traditional biometrics (FaceID, Voice) are compromised. Generative AI can clone a face or voice perfectly. 2,137% growth in deepfake fraud; $40B projected bank losses by 2027; only 24.5% of deepfakes are detectable by humans.
Bio-Pulse Authenticator uses Remote Photoplethysmography (rPPG) to detect a real heartbeat via standard webcam. Every heartbeat causes tiny invisible color shifts in the face. Our AI reads this signal to confirm a living human - not a photo, video, or deepfake.
Zero-Trust Innovation: Shifts from “Is this your face?” to “Is there a human behind this face?”. Works with any standard webcam - no new hardware needed. 4-second UX: Trigger → Verify → Detect → Approve.

Technical Approach

Technologies Used

- Python + OpenCV — captures live webcam video frames for real-time facial video analysis
- MediaPipe Face Mesh — tracks 468 facial landmarks to precisely locate forehead and cheek ROI regions
- open-rppg / pyVHR — open-source rPPG libraries that extract heartbeat signals from subtle skin color changes; no model training needed
- React / Streamlit — renders a live pulsing BPM waveform dashboard with instant ACCESS GRANTED / DENIED result

Implementation Flow

- Step 1 — Trigger: User initiates a high-value transaction (e.g. $5,000+); system opens a Security Scan window
- Step 2 — Capture: Webcam streams live video; MediaPipe maps 468 facial landmarks and isolates forehead/cheek ROI at 30fps
- Step 3 — Signal Extraction: rPPG library reads micro color fluctuations in skin; bandpass filter (0.75–3 Hz) isolates heartbeat frequency from noise
- Step 4 — Visualise: Live BPM waveform drawn over the user’s face in real time; graph pulses green for a real human
- Step 5 — Decision: Valid pulse detected → Liveness Confirmed + Identity Verified → ACCESS GRANTED. Flat-line (photo/video/deepfake) → ACCESS DENIED instantly

Feasibility & Viability

Can We Build This in 12 Hours?

Yes — no model training needed; open-rppg / pyVHR are plug-and-play libraries; the full stack (Python + OpenCV + MediaPipe + React) is free and open-source

Zero new hardware required — works on any standard laptop or smartphone webcam already built into every banking app

Lightweight signal processing pipeline runs in real time on consumer hardware; total latency under 4 seconds end-to-end

Potential Challenges & Mitigations

Lighting sensitivity — rPPG accuracy drops in very low or uneven light; mitigated by adaptive ROI selection and an on-screen lighting guidance prompt

Processing latency on low-end devices — signal processing runs on a background thread to keep the UI responsive; a visible 4-second countdown guides the user

Motion artifacts from head movement — MediaPipe continuously re-tracks landmarks each frame; signal outliers are dropped via the bandpass filter

Spoofing via high-res screen replay — a flat-line or synthetic pulse pattern triggers instant ACCESS DENIED; demo proven live with iPad replay attack

Scalability — stateless API design means Bio-Pulse can be deployed as a microservice and called by any existing banking or fintech app with one API endpoint

Impact and Benefits

Potential Impact on the Target Audience

- Banks/fintech: drop-in liveness layer for high-value transactions, no hardware upgrades
- End users: visible 4-second pulse check builds real trust
- Security teams: unforgeable heartbeat signal defeats deepfakes, photos, and replays
- Compliance: physiological proof-of-presence strengthens KYC/AML audit trails

Benefits of the Solution

- Social: restores digital identity trust; protects vulnerable users from deepfake fraud
- Financial: tackles $40B loss problem; 10% reduction alone saves $4B industry-wide
- Technical: real signal processing (not just an API call); stateless microservice scales to any platform

Research and References

Research Backing

- rPPG Science: De Haan & Jeanne (2013) — "Robust Pulse Rate From Chrominance-Based rPPG" — IEEE Transactions on Biomedical Engineering; foundational paper proving heartbeat can be extracted from facial video using chrominance signals
- MIT Media Lab Study — only 24.5% of high-quality deepfakes are detectable by the human eye; confirms the need for physiological rather than visual liveness checks
- Deloitte / Accenture 2024 Report — generative-AI fraud projected to cost banks $40B by 2027; deepfake fraud attempts grew 2,137% over the last 3 years

Libraries & Tools

- pyVHR Framework: github.com/phuselab/pyVHR — open-source remote PPG library; peer-reviewed, used in academic rPPG benchmarking studies
- open-rppg: github.com/open-rppg/open-rppg — lightweight real-time rPPG implementation optimised for webcam streams
- MediaPipe Face Mesh: ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker — Google's 468-point real-time facial landmark detection
- OpenCV Documentation: docs.opencv.org — video capture and frame processing pipeline
- Claude (Anthropic)